ASSESSING HEART & NECK VESSELS

Chapter 21

Weber

NURS3120: Health Assessment and Promotion

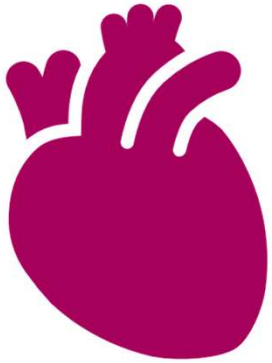


OBJECTIVES

- Discuss the structure and function of the heart and neck vessels
- Discuss the risk factors for coronary artery disease and ways to reduce one's risks
- Differentiate between normal and abnormal findings of the heart and neck vessels

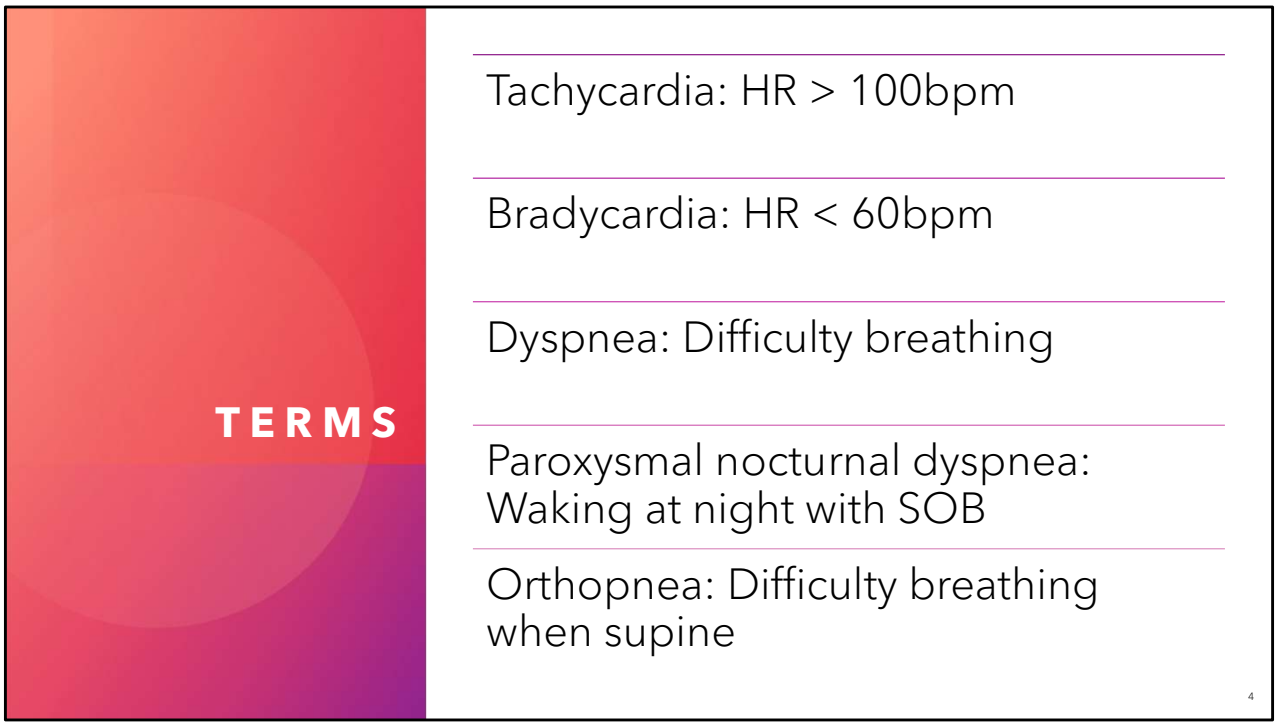
OVERVIEW OF THE CARDIOVASCULAR SYSTEM

- Delivered oxygenated blood to tissues, removes waste products
- Heart pumps blood to all organs & tissues
- Vascular network (arteries & veins) carries blood throughout the body, keeps the heart filled with blood & maintains blood pressure
- Generates an electrical impulse & conducts it through the heart

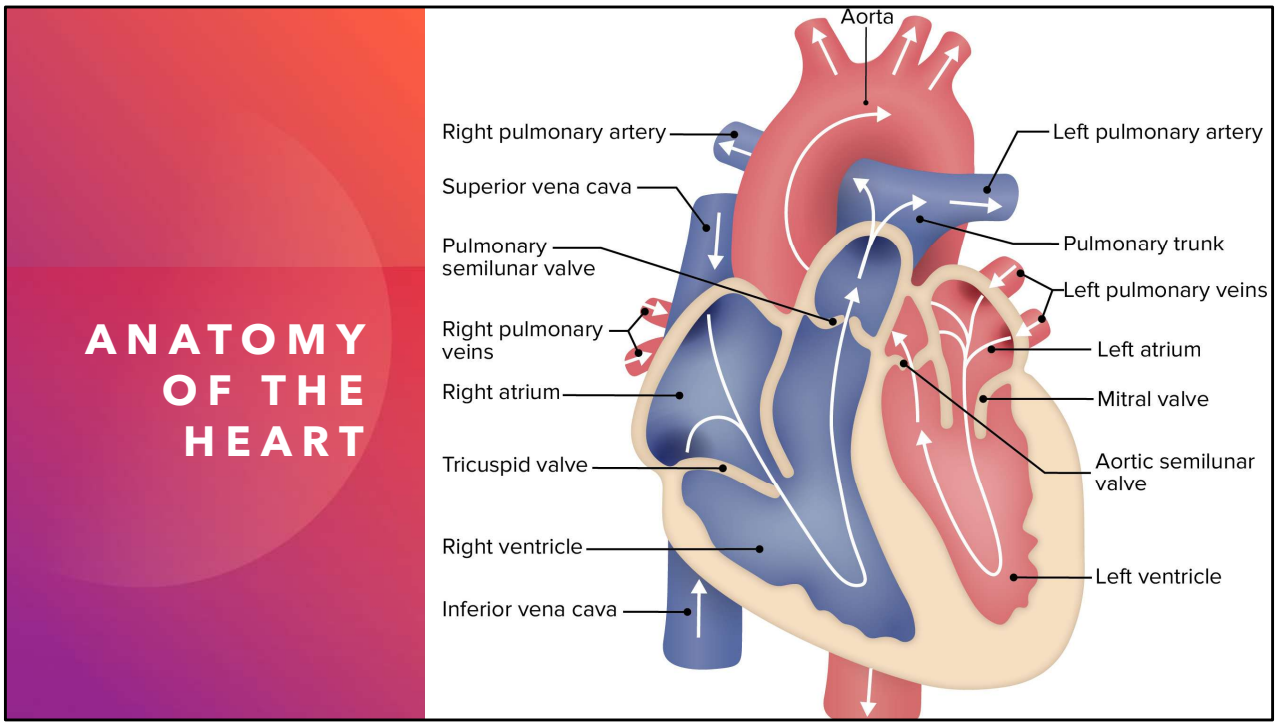


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The cardiovascular system is highly complex, consisting of the heart and a closed system of blood vessels. The heart delivers oxygenated blood to tissues and organs then takes the deoxygenated blood back to the heart and lungs to reoxygenate Arteries carry oxygenated blood; veins carry deoxygenated blood.
The heart is continuously filled with blood….to maintain a blood pressure. The heart also has cardiac muscle cells that can generate an electrical impulse and conduct it through the heart. These electrical impulses regulate the events associated with the filling and emptying of the cardiac chambers. The process is called the cardiac cycle.



Why do you think we are bringing back respiratory terms when talking about the cardiac system?



Great vessels: inferior vena cava, superior vena cava, the aorta, the pulmonary artery, and four pulmonary veins

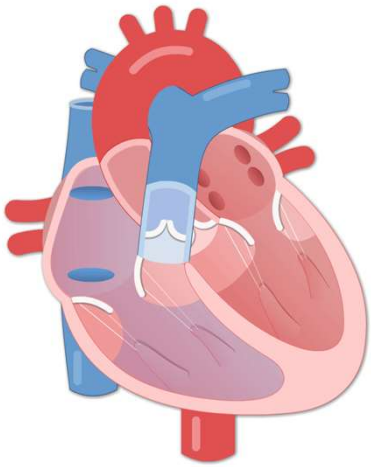
Protected by a thin sac – pericardium. The pericardium is a tough, inextensible, loose-fitting, fibroserous sac that attaches to the great vessels and surrounds the heart. Has an inner layer that forms the epicardium, and an outer layer . The space between the two layers contains 15 to 50 ml of serous fluid, which prevents friction between the layers as the heart pumps.

Four chambers- two atria and two ventricles separated by the cardiac septum. Deoxygenated venous blood returns to the right atrium through three vessels: the superior vena cava, inferior vena cava, and coronary sinus. Blood from the upper body returns to the heart through the superior vena cava. Lower body the inferior vena cava, and blood from the heart muscle itself returns through the coronary sinus. All the blood from those vessels empties into the right atrium.

The anterior chest area that overlies the heart and great vessels is called the precordium.

CARDIAC CYCLE

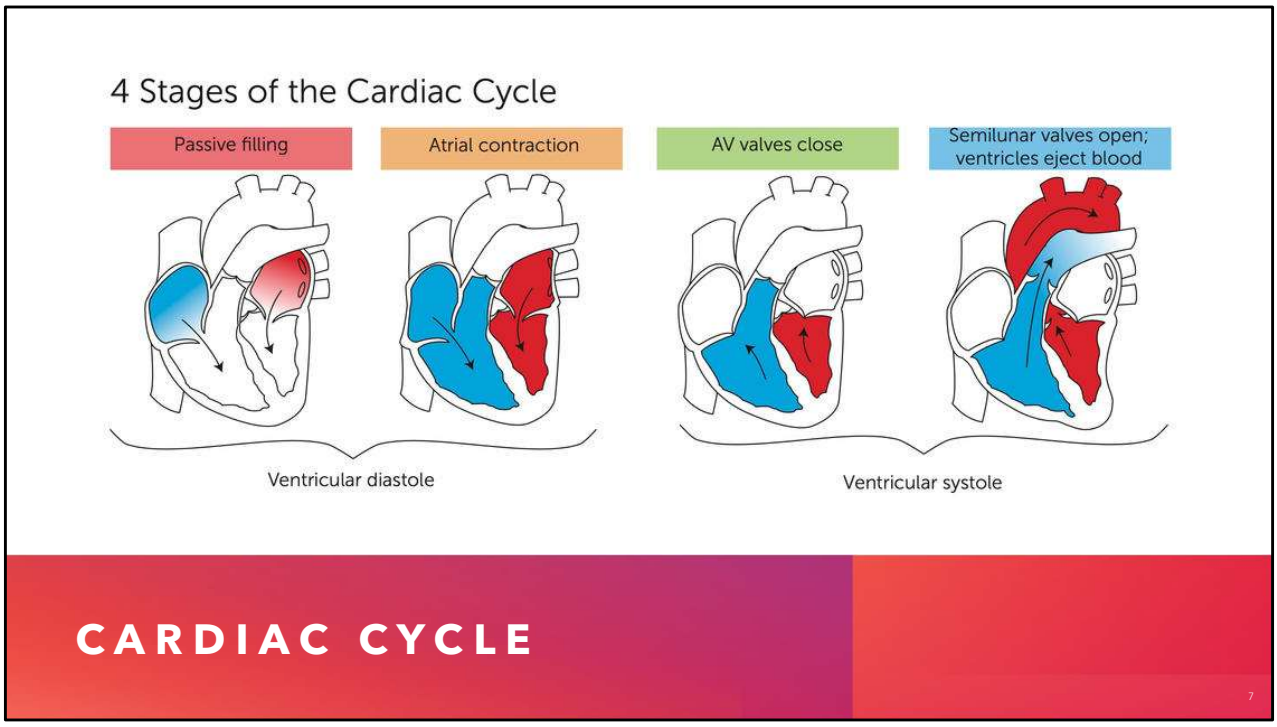
- Refers to the filling & emptying of the heart's chambers
- 2 phases
  - Systole
  - Diastole
- Systole
  - AV valves close, heart contracts & sends blood on its outward journey
- Diastole
  - AV valves open, heart relaxes & fills with blood

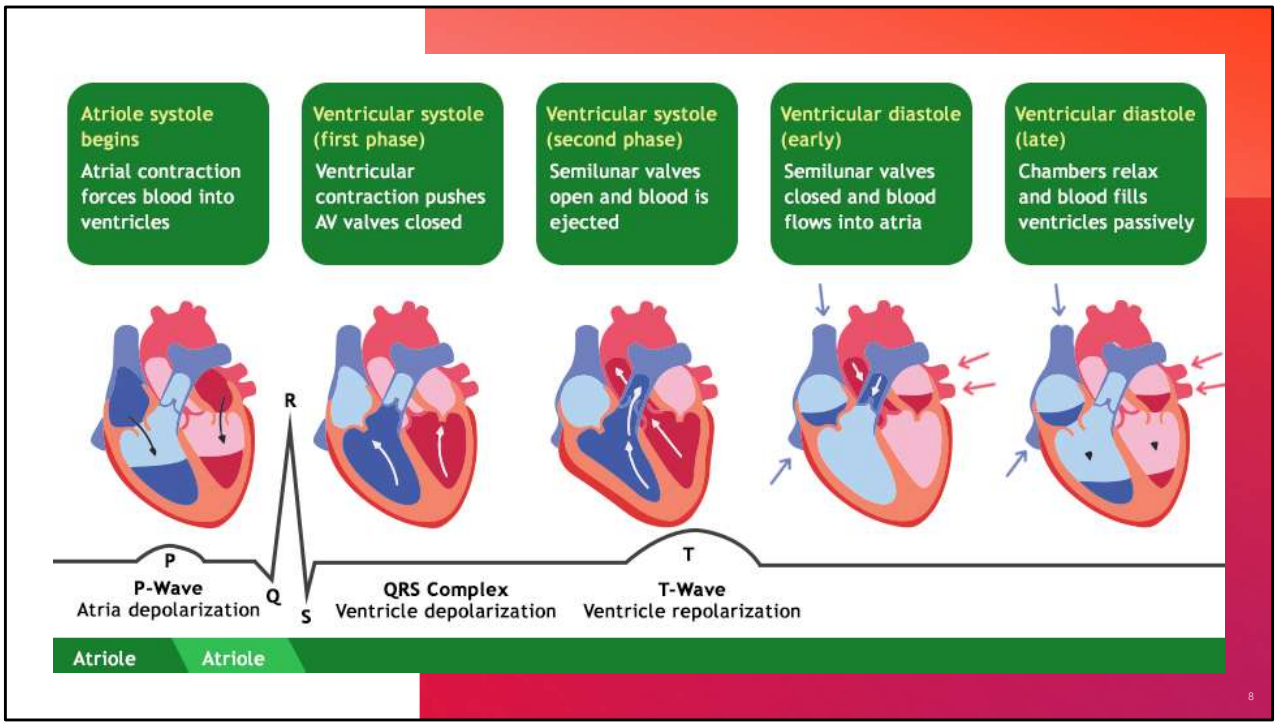


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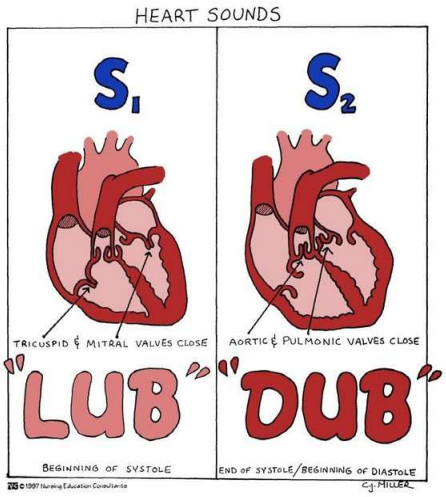
When we talk about the cardiac cycle, we are discussing how the heart is filling and emptying the chambers. diastole (relaxation of the ventricles, known as filling) and systole (contraction of the ventricles, known as emptying).
During diastole, the AV valve (mitral and tricuspid) are open, and the ventricles are relaxed.
The filling phases during diastole result in a large amount of blood in the ventricles, causing the pressure in the ventricles to be higher than in the atria. This causes the AV valves to shut. Closure of the AV valves produces the first heart sound, S1 which is the beginning of systole. The valve closure also prevents blood from flowing backward into the atria during ventricular contraction.
At this point in systole, all four valves are closed and the ventricles contract. There is now high pressure inside the ventricles, causing the aortic valve to open on the left side of the heart and the pulmonic valve to open on the right side of the heart. Blood is ejected rapidly through these valves. With ventricular emptying, the ventricular pressure falls and the semilunar valves close. This closure produces the second heart sound (S2), which signals the end of systole. After closure of the semilunar valves, the ventricles relax. Atrial pressure is now higher than the ventricular pressure, causing the AV valves to open and diastolic filling to begin again.

https://www.youtube.com/watch?v=BEWjOCVEN7M


4 Stages of the Cardiac Cycle
Passive filling
Atrial contraction
AV valves close
Semilunar valves open; ventricles eject blood
Ventricular diastole
Ventricular systole
CARDIAC CYCLE



You will learn about cardiac rhythms later in the program. I just wanted to show you how it correlates with the cardiac cycle.



HEART SOUNDS

- Produced by valve closure
- Normal sounds characterized by "lub-dub"
- Heard with a stethoscope over the precordium
- S1
  - The beginning of systole; best heard at the apex
- S2
  - End of systole/beginning of diastole; best heard at the base

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Heart sounds…when you listen on your client for heart sounds you are hearing the valves close. The opening of valves is silent. Normal heart sounds; "Lub-dub" is S1 and S2.

S1 is the closure of what AV valves (mitral and tricuspid)
S2 is the closure of the semilunar valves (aortic and pulmonic)
Precordium: the area of the anterior chest overlying the heart and great vessels.

COLLECTING SUBJECTIVE DATA

- Chest pain
- Shortness of breath
- Orthopnea
- Edema
- Cyanosis
- Pallor
- History of cardiac disease
- Family history
- Smoke
- Exercise
- Diet
- Medications

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Any chest pain? What does it feel like?, what brought it on? What makes it worse? Did you use some nitroglycerin tabs? Squeezing clenched fist is characteristic of angina (heart blood supply cannot keep up with the metabolic demands). Some individuals may experience chest pain that is not cardiac associated. It can be related to pulmonary, gastrointestinal or musculoskeletal issues.
- these patients may have diaphoresis, cold sweats , pallor, gray appearance, palpitations, dyspnea, nausea, tachycardia, fatigue

Shortness of breath? When does this occur? Paroxysmal nocturnal dyspnea occurs with heart failure, Lying down increases the volume of blood in the chest and the weak heart cannot accommodate the increased load.

Any orthopnea? Ask how may pillows do you sleep on, or they may say they sleep in a recliner or sitting up in a chair.

Cough with sputum production ?

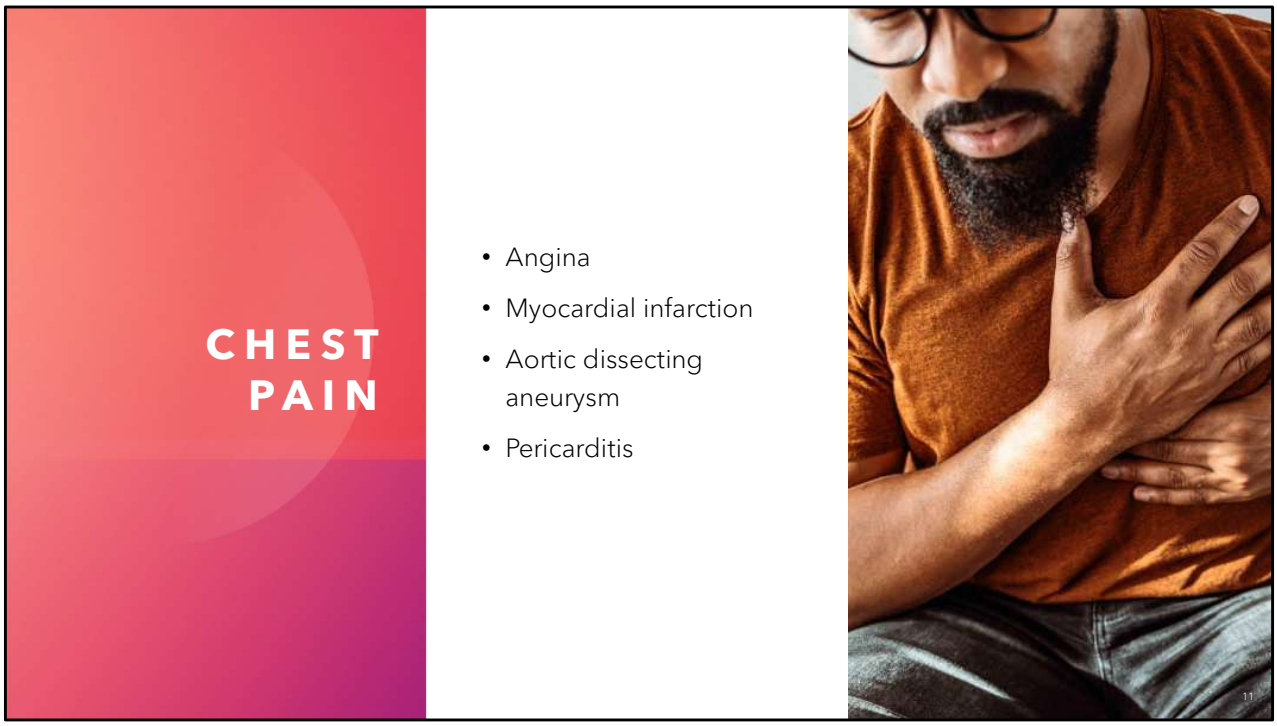
Fatigue – any unusual – is a high symptom in women who have had an MI; fatigue in evening can be from decreased cardiac output – fatigue from anxiety and stress occurs all day or worse in morning.

Cyanosis or pallor occurs with MI or low cardiac output due to decreased tissue perfusion

Any edema? Is it dependent (ankles and feet) with heart failure; cardiac edema is

bilateral where local vein cause will be unilateral.

You want to know any past Hx of cardiac disease, or a family hx. Do you exercise, smoke, consume alcohol, on any meds (antihypertensives, beta-blockers, calcium channel blockers, digoxin, diuretics , ASA, or over the counter)
What are some Risk Factors for heart disease? Age over 65, male, heredity, smoker, HTN, high cholesterol, high triglycerides, physical inactivity, obesity, overweight, DM, excessive alcohol consumption, diet and nutrition, stress



We need to use specific COLDSPA descriptors to discuss and differentiate between the types of chest pain.

Angina (Myocardial ischemia): Tight pressing, heavy, discomfort occurring often with exertion, across the chest and may radiate to the shoulders, jaws, neck or upper abdomen, lasts up to 20 minutes, mild to moderate pain, may be relieved with nitroglycerine, may occur with difficulty breathing, nausea, or sweating

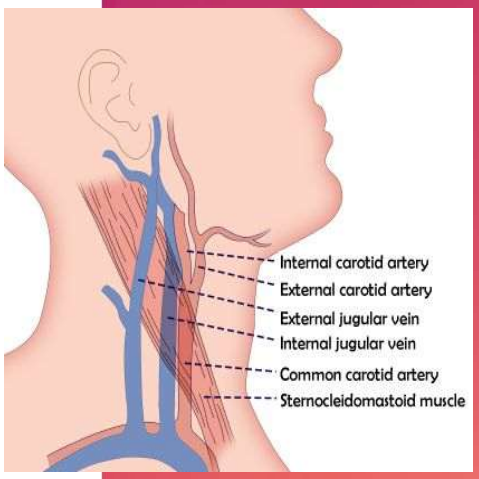
MI (Irreversible heart damage due to myocardial ischemia): heavy tight pressing, discomfort occurring with or without exertion, across the chest and may radiate to shoulders, jaws, neck or upper abdomen, lasts 20 minutes to hours, often moderate to severe pain, rest does not relieve, may occur with nausea, vomiting, weakness and sweating

Aortic dissection (Tear in the wall lining the aorta): tearing, ripping sensation, abrupt, in chest radiating to neck, back or abdomen, lasting for hours, severe, aggravated by hypertension; relieved only with treatment

Pericarditis (Inflammation of the parietal pleura next to the pericardium): sharp, stabbing, pounding feeling when heart beats, ongoing, pericardial pain radiating to shoulders or neck, ongoing, sever, aggravated by breathing position changes, coughing, and may be relieved with sitting forward, may also be seen with fever, weakness, tiredness, coughing, trouble breathing and or pain when swallowing.

OVERVIEW OF THE NECK VESSELS

- Carotid artery
  - Located in the groove between the trachea & the sternomastoid muscle
- Internal & external jugular veins
  - Internal: Deeper in the neck in the sternocleidomastoid muscle
  - External: More superficial lying lateral to the sternomastoid muscle above the clavicle



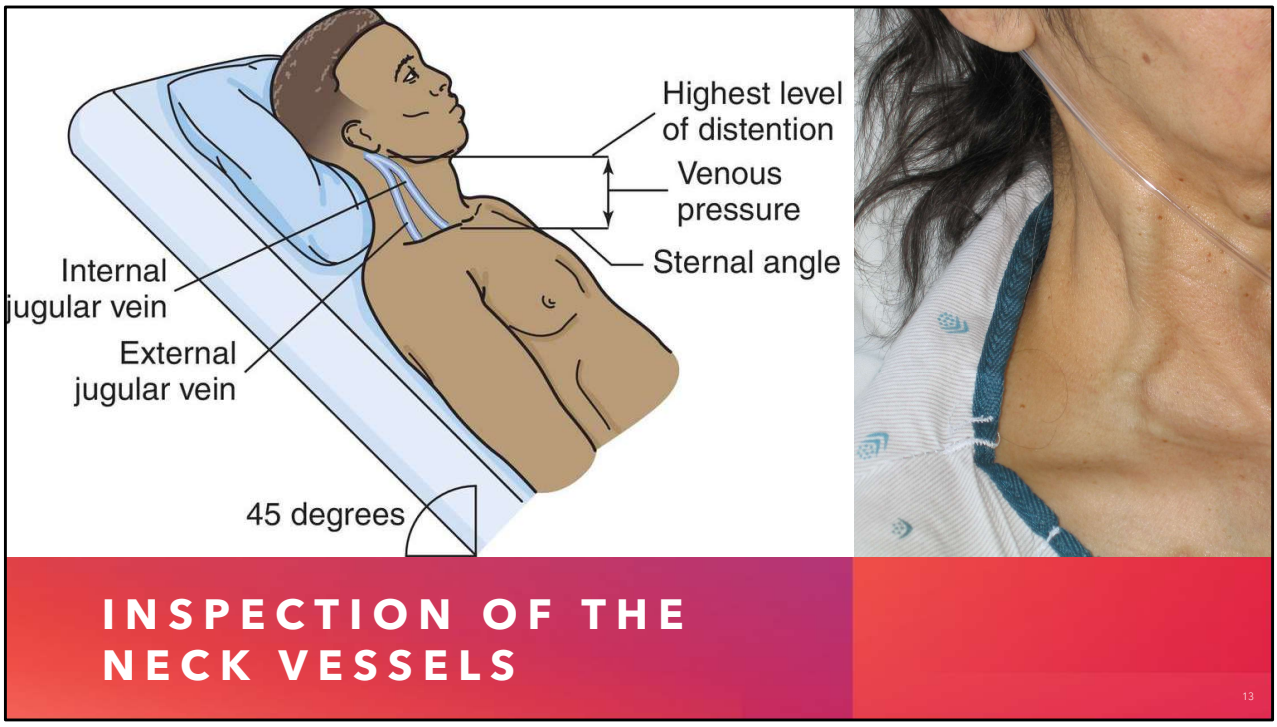
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There are two neck vessels upon assessment of the cardiovascular system that are important. The carotid artery and the internal Jugular vein. Assessment of the pulses of these vessels reflects the integrity of the heart muscle.
The right and left common carotid arteries extend from the brachiocephalic trunk and the aortic arch and are located in the groove between the trachea and the right and left sternocleidomastoid muscles. They are supplying the head and neck including the brain, with oxygenated blood. The carotid pulse is a centrally located arterial pulse.

There are two sets of jugular veins: internal and external. The internal jugular veins lie deep and medial to the sternocleidomastoid muscle. The external jugular veins are more superficial; they lie lateral to the sternocleidomastoid muscle and above the clavicle. The jugular veins return blood to the heart from the head and neck by the way of the superior vena cava.
Assessment of the jugular venous pulse is important for determining the hemodynamics of the right side of the heart. Right-sided heart failure raises pressure and volume, thus raising jugular venous pressure.

- **Evaluation of the neck vessels is an important component of cardiac assessment.**

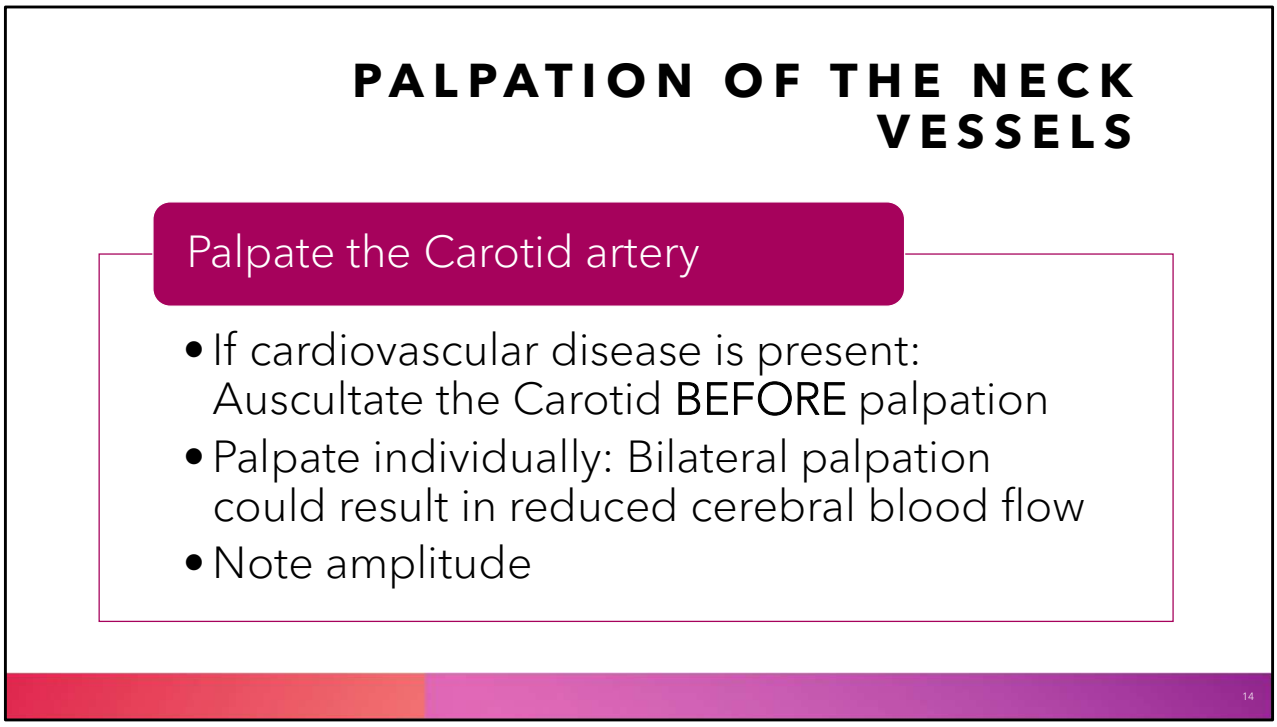
- **They reflect the integrity of the heart.**



Observe the **jugular venous pulse**– have the client lie down, elevate the HOB 30-45 degrees, turn the head away from you. Jugular pulse should not be seen higher than the sternal angle. If seen higher that is an indication of right ventricular failure, pulmonary hypertension, pulmonary emboli, or cardiac tamponade (a life-threatening, emergency condition where fluid, blood, or pus builds up in the sac around the heart (pericardium), preventing chambers from filling properly and causing a drastic drop in blood pressure).

Jugular venous pressure is often omitted and replaced by a medical order of a pulmonary artery catheter placement…. But as nurses we can still evaluate jugular vein distention, position the client in a supine position with the head of the bed elevated 30, then 45, then 60, and then 90 degrees. At each increase of the elevation, have the client's head turned slightly away from the side being evaluated. Using tangential lighting (light that strikes the surface at a low angle), observe for distention, protrusion, or bulging. This may indicate **right sided heart failure**.

- https://www.youtube.com/watch?v=MZKSkVSbH8k



Auscultate if the client is middle-aged or older or if you suspect cardiovascular disease. Auscultate the carotid arteries before you palpate because palpation may increase or slow the HR, changing the strength of the carotid impulse heard.

Auscultate – for those with signs of cardiovascular disease for the presence of a bruit (sound made as a result of turbulent blood flow)
Auscultate: Place the bell of the stethoscope over the carotid artery and ask the client to hold their breath for a moment so that breath sounds do not conceal any vascular sounds. Normal finding: no blowing or swishing or other sounds are heard. A bruit is abnormal: Bruit- caused by narrowing of the carotid artery due to plaque buildup or arterial disease

Palpate the carotid artery: by placing the pads of the index and middle fingers medial to the sternocleidomastoid muscle on the neck. When you are palpating, notice if you feel a thrill……(which feel similar to a purring cat). When you are palpating the carotid arteries note amplitude.
Weak pulses may indicate hypovolemia, shock, or decreased cardiac output. A bounding, firm pulse may indicate hypervolemia or increased cardiac output.
A loss of elasticity may indicate arteriosclerosis.

Thrills may indicate a narrowing of the artery.

AUSCULTATION OF THE CAROTID

- Assess with known/suspected cardiovascular disease
- Use the bell of stethoscope
- Listen over each artery
- Should not hear bruit



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- Use the bell of the stethoscope when auscultating the Carotid
- Listen over each artery
- Shouldn't hear sounds over the carotid artery.
- A hum or bruit sounds like buzzing or blowing – could indicate arteriosclerotic plaque formation

If you detect occlusion during auscultation, palpate very lightly to avoid blocking circulation or triggering vagal stimulation and bradycardia, hypotension, or even cardiac arrest.

INSPECTION & PALPATION OF THE CHEST

- Inspect
  - General appearance
  - Pulsations on anterior chest
- Palpate
  - Apical pulse
  - Abnormal pulsations



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Inspect the client when the client is lying in the supine position with the head of the bed about 30-45 degrees. Stand on the client's right side and note for any abnormal pulsations. The apical impulse may or may not be visible. The apical pulse is a result of the left ventricle moving outward during systole. Pulsations, which are also known as heaves or lifts, are considered abnormal and should be evaluated. A heave or lift may occur as the result of an enlarged ventricle from an overload of work.

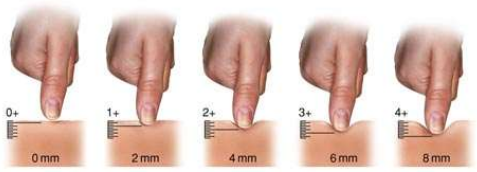
Palpate the apical impulse: Have the client lie supine. Use one or two finger pads to palpate the apical impulse in the mitral area (fourth or fifth ICS at the MCL). If the pulsation can not be palpated, then have the client assume a left lateral position. This displaces the heart toward the left chest wall and relocates the apical impulse farther to the left.
Normal findings is palpated may be the size of a nickel (1-2cm). Amplitude is usually small (like a small tap), and duration is brief.
Abnormal findings is when the apical impulse is larger than 1-2cm, displaced, more forceful, or of longer duration, suspect cardiac enlargement.
In older clients, the apical impulse may be difficult to palpate because of increased anteroposterior chest diameter. The apical impulse may be impossible to palpate in clients with pulmonary emphysema. The apical impulse may not be palpable in

clients who have obesity or have large breasts.

Palpate for abnormal pulsations: using your palmar surface of your hand, palpate the apex, left sternal border and base. Normal: no pulsations or vibrations are palpated. Abnormal: a thrill or a pulsation is usually associated with a grade 4 or higher murmur.

PALPATING EDEMA

- Graded on a four-point scale
- 0+ - 4+



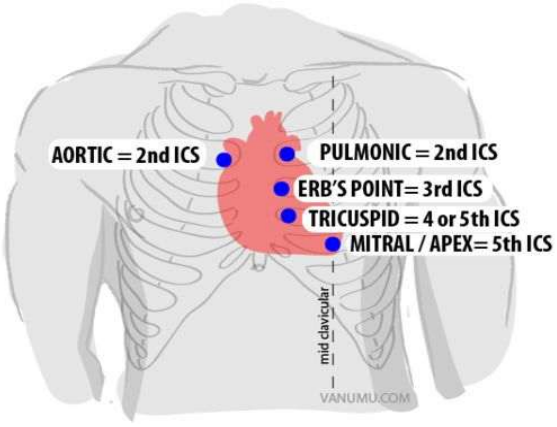
0+ No pitting edema
1+ Mild pitting edema. 2 mm depression that disappears rapidly.
2+ Moderate pitting edema. 4 mm depression that disappears in 10–15 seconds.
3+ Moderately severe pitting edema. 6 mm depression that may last more than 1 minute.
4+ Severe pitting edema. 8 mm depression that can last more than 2 minutes.

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Assessment of edema is part of the peripheral vascular system, but it also clues us to the functioning of the cardiovascular system. Bilateral edema in both lower extremities at night is seen in heart failure due to a reduction of blood flow out of the heart, causing blood returning to the heart to back up in the organs and dependent areas of the body.

When we grade edema: 0 would be no pitting edema, 1+ mild pitting edema 2mm depression that disappears rapidly, 2+ moderate pitting edema 4mm depression usually comes back after 10-15 seconds, 3+ moderately severe pitting edema 6mm depression, rebounds back after 1 minute and 4+ edema is severe pitting edema 8mm depression and takes longer than 2 minutes to rebound back

If you have a heart failure client, how do you think their edema is? - edema is bilateral, worse at night

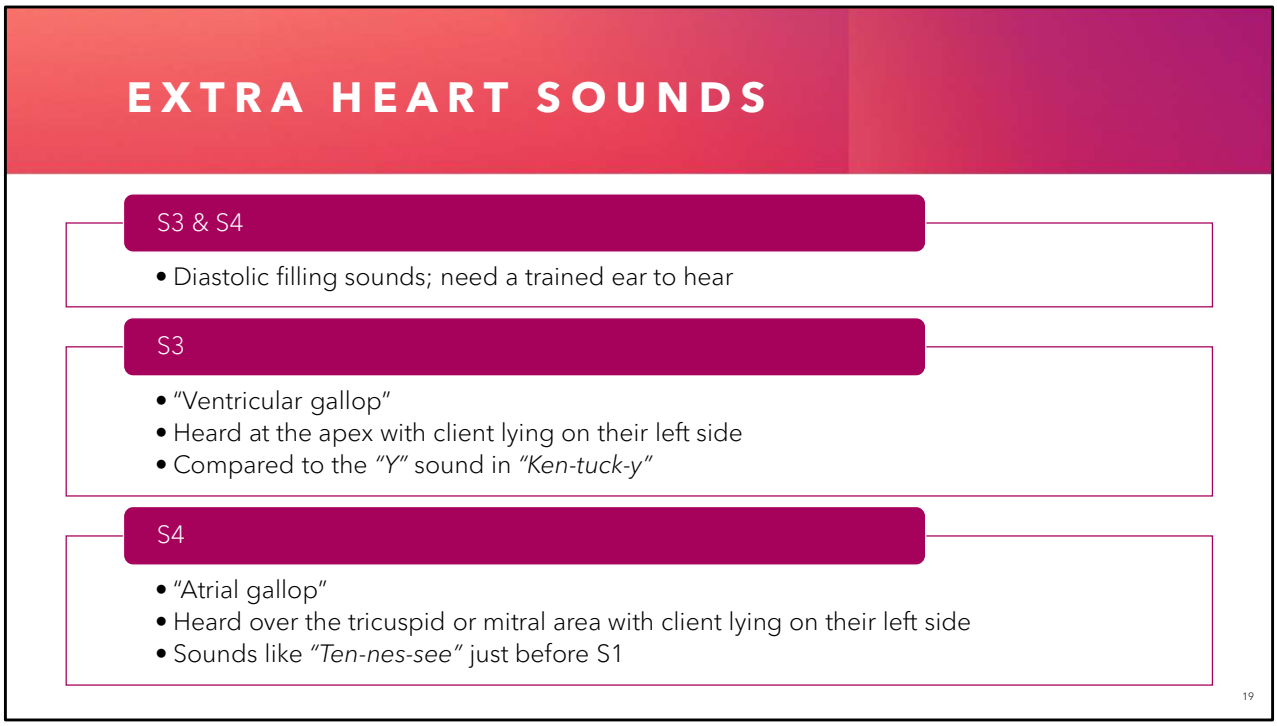


AUSCULTATION OF THE HEART

Auscultate HR and Rhythm: Place the diaphragm of the stethoscope at the apex and listen closely to the rate and rhythm. Normal: Rate 60-100 beats/min and regular. When a client has irregular rhythms, it predisposes the client to decreased CO, heart failure, or emboli

Auscultate to identify S1 S2: Auscultate the first heart sound and the second heart sound. Lub Dub: S1 starts systole and S2 starts diastole. S1 corresponds with each carotid pulsation and is loudest at the apex of the heart. S2 immediately follows after S1 and is loudest at the base of the heart.

Use **both** your diaphragm and bell to listen to the heart sounds
Aortic: 2nd ICS right sternal border
Pulmonic: 2nd ICS at the left sternal border
Erb point: 3rd ICS at the left sternal border
Tricuspid: Fourth or fifth ICS at the left lower sternal border
Mitral: Fifth ICS near the left MCL-the apex of the heart



Extra sounds are a result from ventricular vibration secondary to rapid ventricular filling. If present, S3 can be heard early in diastole, after S2; often termed ventricular gallop. To attempt to listen to S3, the best way is to place your client on their left side and listen at the apex of the heart….

https://www.youtube.com/watch?v=Sms-US-bncg

S4 can be heard late in diastole, just before S1 and is often termed atrial gallop. Best to hear over the tricuspid or mitral area. You can hear the and S4 in the elderly, HTN, aortic stenosis, or a history of an MI. It sounds like "Ten-nes-see" just before the S1 after atrial contraction.

MURMURS

- Sound results from turbulent blood flow: Results in swooshing or blowing sound
- Can occur during systole or diastole
- Graded from 1-6: Barely audible to loud without a stethoscope on the chest, associated thrill
- Best way to auscultate is with the client sitting up & leaning forward

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Normally blood flows silently through the heart, but sometimes there are conditions that can create turbulent blood flow in which a swooshing or blowing sound may be auscultated. Increased blood velocity, structural valve defects, valve malfunctions and abnormal chamber openings (septal defect). Murmurs can occur during systole or diastole

It is very important to determine where the murmur occurs; a systolic murmur can be present in a healthy heart whereas a diastolic murmur always indicates heart disease.



Murmur characteristics: Pitch: murmurs can assume a high, medium or low pitch
Intensity: Murmur grade scale 1-6
Quality: The sound murmurs make has been described as blowing, rushing, roaring, rumbling, harsh, or musical
Shape/Pattern: The shape of a murmur is determined by its intensity from beginning to end. There are four different categories of shape: crescendo (growing louder), decrescendo (growing softer), crescendo-decrescendo (growing louder and then growing softer) and plateau (staying the same throughout)
Timing: Is the murmur systolic versus diastolic

GRADING MURMURS

- **Grade I:** Barely audible
- **Grade II:** Audible but quiet and soft
- **Grade III:** Moderately loud, without thrust or thrill
- **Grade IV:** Loud, with a thrill
- **Grade V:** Very loud, heard with a stethoscope only partially in contact with the chest wall with a heave or thrill
- **Grade VI:** Loud enough to be heard before the stethoscope contacts the chest wall with a thrill

Cardiac Murmur	
Systolic Murmur	**Diastolic Murmur**
Aortic stenosis	Mitral stenosis
Mitral regurgitation	Aortic regurgitation
Mitral valve prolapse	
Tricuspid regurgitation	

HEALTH PROMOTION AND DISEASE PREVENTION: CORONARY HEART DISEASE

- Broad category with a range of diseases
  - Coronary artery disease, diseases of blood vessels, heart rhythm problems, heart infections, and congenital heart defects
- Cardiovascular disease and coronary heart disease used interchangeably with heart disease
  - Refers to conditions involving narrowed or blocked blood vessels, thos that affect heart muscles, valves, or rhythm
- Leading cause of death in the US

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Although the complexity of how heart disease develops is still being studied, it is well recognized that lifestyle affects the disease and healthy changes in lifestyle may reduce or reverse vascular changes leading to the disease

CORONARY HEART DISEASE

- Healthy People 20230 Goal:
  - Improve cardiovascular health and quality of life through prevention, detection, and treatment of risk factors for heart attack and stroke; early identification and treatment of heart attacks and strokes; and prevention of repeat cardiovascular events.
- Managing risk is key

CORONARY HEART DISEASE

- Increasing age; especially over age 65
- Sex assigned at birth
- >45 in males; >55 in females
- Heredity
- Smoking
- High blood cholesterol
- Hypertension
- Physical activity
- Overweight/obesity
- Diabetes mellitus
- Stress
- Excessive alcohol intake
- Poor diet & nutrition

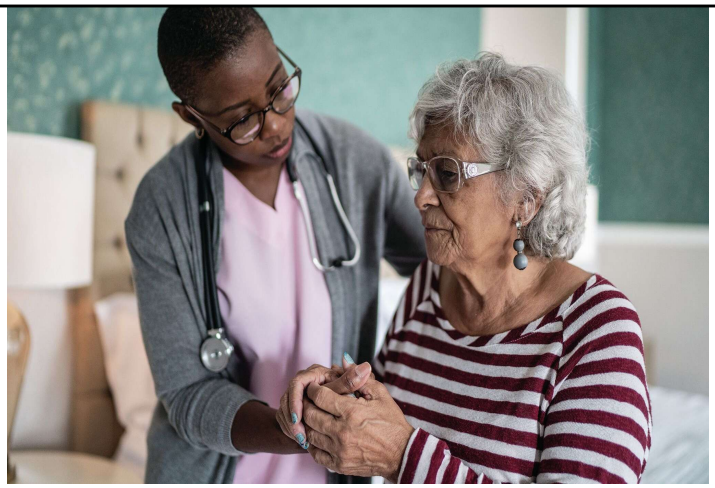
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Screening for risk of heart disease includes blood tests for cholesterol level, glucose level, and presence of C-reactive protein, blood pressure measurement, a health history assessing cardiovascular-related risks, and screening for peripheral artery disease.

Education!

- Stop smoking
- Healthy diet
- Reduce elevated cholesterol (diet, activity, medication)
- Lower BP (weight loss, increase activity, medications)
- Increase physical activity
- Maintain healthy weight
- Manage diabetes
- Limit alcohol intake to 1-2 per day for males and one per day for females
- Stress reduction

- Slight decrease in heart size
- Loss of cardiac contractile strength & efficiency
- Decreased cardiac output (30-35% by age 70)
- Thickening of the heart valves, causing incomplete valve closure



OLDER ADULT CONSIDERATIONS

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Cardiovascular changes with aging: A client can also have an enlarged heart due to cardiovascular disease like CAD, HTN

OLDER ADULT CONSIDERATIONS

- Lifestyle influences the heart, not just the aging process
- Orthostatic hypotension: Sudden drop in BP when rising to sit or stand
- AP diameter may increase, causing difficulty in feeling the apical impulse
- Ectopic beats are common but do not indicate heart disease; obtain an ECG for clarification



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Poor health habits contribute greatly to cardiac disease
Smoking, obesity, diabetes, high cholesterol, lack of exercise/sedentary lifestyle, genetics
Need to instruct the individual not to get up suddenly from a lying or seated position to prevent orthostasis



OLDER ADULT CONSIDERATIONS

- Fibrous tissue infiltration of sinoatrial node & intermodal atrial tracts, causing arrhythmias
  - Primarily atrial fibrillation & premature atrial & ventricular contractions
- Increased aortic rigidity
- Increased amount of time necessary for the heart rate to return to normal after exercise
- Decreased strength & elasticity of blood vessels, contributing to arterial & venous insufficiency

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Chest x-ray shows a very enlarged heart with increased pulmonary vascular outline indicative of congestive heart failure

OLDER ADULT CONSIDERATIONS



Decreased ability to respond to physical & emotional stress



Increased peripheral vascular resistance, leading to increased blood pressure



Decreased arterial wall elasticity, leading to isolated systolic hypertension

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QUESTIONS?



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